Cairo University	MTHG001 (Algebra)	Fall 2024
Faculty of Engineering	Midterm Exam	Duration: 60 minutes

ظلل إجاباتك بوضوح في الدوائر المخصصة فقط. حافظ على نظافة ورقتك.

Shade your answers clearly in the bubble sheet only. Keep your paper clean.

1. A square matrix which is not row equivalent to the identity matrix, I, is:
   (a) singular (b) rectangular (c) nonsingular (d) elementary (e) None

Consider $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

2. The elementary row operation performed on a 3×4 matrix when it is multiplied from the left by E is:
   (a) $-3r_3 + r_2 \to r_2$ (b) $-3r_2 + r_3 \to r_3$
   (c) $3r_3 + r_2 \to r_2$ (d) $3r_2 + r_3 \to r_3$ (e) None
3. The entry in row 2 and column 3 of E^{-1} is:
   (a) 3 (b) 1/3 (c) $-1/3$ (d) -3 (e) None
4. Consider the linear system $A\underline{X} = \underline{B}$. If the size of A is $m \times n$, where $m > n$, and $\underline{B} \neq \underline{0}$, then the system cannot have:
   (a) no solution (b) infinitely many solutions
   (c) the trivial solution (d) None
5. If the matrix $\begin{bmatrix} 1 & 3 & f^2-4 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & f+2 & 2 \end{bmatrix}$ is in row echelon form (REF), then f is equal to:
   (a) 2 (b) -1 (c) -3 (d) -2 (e) None
6. If A is an $n \times n$ skew-symmetric nonsingular matrix, then $(A^{-1})^6$ is:
   (a) skew-symmetric (b) diagonal
   (c) symmetric (d) triangular (e) None
7. If A, B, C, D, and F are $n \times n$ nonsingular matrices, and $DAXBC = F$, then X is equal to:
   (a) $A^{-1}D^{-1}FC^{-1}B^{-1}$ (b) $D^{-1}A^{-1}FC^{-1}B^{-1}$
   (c) $A^{-1}D^{-1}FB^{-1}C^{-1}$ (d) $D^{-1}A^{-1}FB^{-1}C^{-1}$ (e) None
8. D, L, and U are $n \times n$ nonsingular matrices. D is diagonal, L is lower triangular, and U is upper triangular. $(UD^2)^{-1}L^T$ is:
   (a) lower triangular (b) upper triangular
   (c) symmetric (d) diagonal (e) None

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9. If $(A^T - 0.3I)^{-1} = \begin{bmatrix} 1 & 3 \\ 2 & g \end{bmatrix}$, $g \neq 6$, then the entry in row 1 and column 2 of A is equal to:

(a) $2/(g-6)$ (b) $-3/(g-6)$

(c) $3/(g-6)$ (d) $-2/(g-6)$ (e) None

Consider the linear system $\begin{aligned} x + y + 3z &= 2 \\ 4x + 5y + 2z &= -4 \\ 3x + y + 29z &= 30 \end{aligned}$. If $z = t, t \in \mathbb{R}$, then:

10. y is equal to:

(a) $-12 - 10t$ (b) $10 + 12t$ (c) $-12 + 10t$ (d) $10 - 12t$ (e) None

11. x is equal to:

(a) $14 + 13t$ (b) $14 - 13t$ (c) $13 + 14t$ (d) $13 - 14t$ (e) None

12. If $A = \begin{bmatrix} 1 & -3 \\ r & 2 \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 2 and column 1 of $A^T A^2$ is equal to:

(a) $12r - 12$ (b) $7r - 8$ (c) $10r - 2$ (d) $15r - 3$ (e) None

Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 9 \\ -6 & -2 & k \end{bmatrix}$, where $|k| < 8$.

13. The entry in row 2 and column 3 of U is equal to:

(a) -1 (b) 3 (c) 4 (d) 5 (e) None

14. The entry in row 3 and column 3 of L is equal to:

(a) $k + 10$ (b) $1/(k+10)$ (c) $k + 6$ (d) $1/(k+6)$ (e) None

15. The linear system $\begin{aligned} x + 2y + 3z &= a \\ 3x - y + 4z &= b \\ 4x + 22y + 22z &= c \end{aligned}$, where $a, b, c \in \mathbb{R}$, will be consistent if:

(a) $c = -5a + 3b$ (b) $c = -2a + 2b$

(c) $c = a + b$ (d) $c = 10a - 2b$ (e) None

16. Consider the linear system $\begin{aligned} x + y - z &= a \\ 2x + 5y + 2z &= -2 \\ 3x + 2y + z &= 2 \end{aligned}$. If $x = r, y = s$, and $z = 3$ is the solution of this system, where $r, s, a \in \mathbb{R}$, then a is equal to:

(a) -4 (b) 4 (c) 1 (d) -1 (e) None

17. Consider the LU decomposition of the matrix $\begin{bmatrix} 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 5 \\ -4 & 2 & 4 & 6 \\ 6 & 4 & 1 & 4 \end{bmatrix}$. The entry in row 4 and column 4 of L is equal to:

(a) 1/27 (b) 28 (c) 27 (d) 1/28 (e) None

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Consider $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 39 \\ 2 & 5 & 5 & 6 & 2 & 2 \end{bmatrix}$. Let R be the reduced row echelon form (RREF) of A.

18. The entry in row 2 and column 6 of R is equal to:
    (a) -53 (b) -50 (c) -47 (d) -44 (e) None

19. The entry in row 1 and column 6 of R is equal to:
    (a) 81 (b) 65 (c) 57 (d) 73 (e) None

20. If A and B are $n \times n$ nonsingular matrices, then $A^2(A^{-1} + B^{-1})B(A + B)^{-1}B$ is equal to:
    (a) AB^2 (b) BA (c) AB (d) BA^2 (e) None

21. The linear system $\begin{aligned} 2x + ay - z &= 0 \\ -3x + by + 3z &= 0 \\ x + cy - 2z &= 0 \end{aligned}$, where $a, b, c \in \mathbb{R}$, will have nontrivial solutions if:
    (a) $a + b + c = 0$ (b) $4a + 3b + c = 0$
    (c) $2a + 3b + 5c = 0$ (d) $a + 3b + 7c = 0$ (e) None

Consider the linear system $\begin{bmatrix} 1 & -2 & 4 \\ 2 & -3 & 11 \\ 3 & -3 & 21 \\ 4 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$, where $a, b, c, d \in \mathbb{R}$.

22. The system will be consistent if:
    (a) $c = 4b - 5a$ (b) $c = 3b - 3a$
    (c) $c = 5b - 7a$ (d) $c = 2b - a$ (e) None

23. If the system has a unique solution, and $z = 1$, then:
    (a) $d = 9b - 14a - 40$ (b) $d = 9b - 14a - 39$
    (c) $d = 12b - 20a - 51$ (d) $d = 11b - 18a - 47$ (e) None

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1. (a) Singular

2. (c) $3r_3 + r_2 \longrightarrow r_2$

3. (d) -3

4. (c) the trivial solution $(B \neq 0)$

5. (b) $f + 2 = 1 \therefore f = -1$ (b)

6. (b) $A^{-1} \rightarrow$ skew $\therefore (A^{-1})^6 \rightarrow$ diagonal (b)

7. (a) $X = A^{-1}D^{-1}FC^{-1}B^{-1}$

8. (b) $D^2 \rightarrow$ diagonal $\therefore UD^2 \rightarrow$ upper $\therefore (UD^2)^{-1} \rightarrow$ upper $\therefore (UD)^{-1}L^T \rightarrow$ upper (b)
(upper)

9. (d) $A = \left[\begin{bmatrix} 1 & 3 \\ 2 & q \end{bmatrix}^{-1} + 0.3I\right]^T = \left[\begin{bmatrix} \frac{q}{q-6} & \frac{-3}{q-6} \\ \frac{-2}{q-6} & \frac{1}{q-6} \end{bmatrix} + \begin{bmatrix} 0.3 & 0 \\ 0 & 0.3 \end{bmatrix}\right]^T = \begin{bmatrix} \frac{q}{q-6} + 0.3 & \frac{-2}{q-6} \\ \frac{-3}{q-6} & \frac{1}{q-6} + 0.3 \end{bmatrix}$

10. (c) $x + y + 3t = 2$, $4x + 5y + 2t = -4 \therefore y = -12 + 10t$ (c)

11. (b) $x + (-12 + 10t) + 3t = 2 \therefore x = -13t + 14$ (b)

12. (d) $A^TA^2 = \begin{bmatrix} 1 & r \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 1 & -3 \\ r & 2 \end{bmatrix}\begin{bmatrix} 1 & -3 \\ r & 2 \end{bmatrix} = \begin{bmatrix} 1+r^2 & -3+2r \\ -3+2r & 13 \end{bmatrix}\begin{bmatrix} 1 & -3 \\ r & 2 \end{bmatrix} = \begin{bmatrix} 3r^3 - 3r + 1 & -3r^2 + 4r - 9 \\ 15r - 3 & -6r + 35 \end{bmatrix}$

13. (a)
$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 9 \\ -6 & -2 & K \end{bmatrix} \xrightarrow{\frac{1}{2}r_1 \rightarrow r_1} \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ 4 & -1 & 9 \\ -6 & -2 & K \end{bmatrix} \xrightarrow[6r_1 + r_3 \rightarrow r_3]{-4r_1 + r_2 \rightarrow r_2} \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & -3 & 3 \\ 0 & 1 & K+9 \end{bmatrix} \xrightarrow{\frac{r_2}{-3} \rightarrow r_2} \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & -1 \\ 0 & 1 & K+9 \end{bmatrix}$$

$$\xrightarrow{r_1 + \frac{1}{2}r_2 \rightarrow r_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & K+10 \end{bmatrix} \xrightarrow{\frac{r_3}{K+10} \rightarrow r_3} \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

U

$$\begin{bmatrix} 2 & 0 & 0 \\ 4 & -3 & 0 \\ -6 & 1 & K+10 \end{bmatrix}$$

L

14. (a)

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15. (d) $\left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 3 & -1 & 4 & b \\ 4 & 22 & 22 & c \end{array}\right) \xrightarrow[-4r_1+r_3\to r_3]{-3r_1+r_2\to r_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & -7 & -5 & b-3a \\ 0 & 14 & 10 & c-4a \end{array}\right) \xrightarrow{\frac{r_2}{-7}\to r_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & \frac{5}{7} & \frac{b-3a}{-7} \\ 0 & 14 & 10 & c-4a \end{array}\right)$

$\xrightarrow{-14r_2+r_3\to r_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & \frac{5}{7} & \frac{b-3a}{-7} \\ 0 & 0 & 0 & c-4a+2(b-3a) \end{array}\right)$ $\therefore c-4a+2b-6a=0 \therefore \boxed{c=10a-2b}$ (d)

16. (a) $2x+5y+2(3)=-2$, $3x+2y+3=2 \therefore x=1, y=-2 \therefore \boxed{a=1-2-3=-4}$ (a)

17. (c) $\begin{bmatrix} 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 5 \\ -4 & 2 & 4 & 6 \\ 6 & 4 & 1 & 4 \end{bmatrix} \xrightarrow{\frac{r_1}{4}\to r_1} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 2 & 1 & 3 & 5 \\ -4 & 2 & 4 & 6 \\ 6 & 4 & 1 & 4 \end{bmatrix} \xrightarrow[4r_1+r_2\to r_2]{-2r_1+r_2\to r_2} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 6 & 12 \\ 0 & 4 & -2 & -5 \end{bmatrix}$

$\xrightarrow[-6r_1+r_3\to r_3]{}$

$\xrightarrow{-2r_2+r_3\to r_3} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -10 & -13 \end{bmatrix} \xrightarrow[4r_3+r_4\to r_4]{} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 27 \end{bmatrix}$

$\left(\frac{r_3}{2}\to r_3\right)$ $\left(\frac{r_4}{27}\to r_4\right)$

$\begin{bmatrix} 4 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -4 & 2 & 2 & 0 \\ 6 & 4 & -10 & 27 \end{bmatrix}$

4 L

18. (b) $\begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 39 \\ 2 & 5 & 5 & 6 & 2 & 2 \end{bmatrix} \xrightarrow[-4r_1+r_3\to r_3]{-3r_1+r_2\to r_2} \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 0 & 1 & -1 & 2 & 3 & 4 \\ 0 & 2 & -2 & 5 & 5 & 15 \\ 0 & -1 & 1 & -4 & 0 & -10 \end{bmatrix} \xrightarrow[-2r_2+r_3\to r_3]{-3r_2+r_1\to r_1} \begin{bmatrix} 1 & 0 & 5 & -1 & -8 & -6 \\ 0 & 1 & -1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & -1 & 7 \\ 0 & 0 & 0 & -2 & 3 & -6 \end{bmatrix}$

19. (d)

$\xrightarrow[-2r_3+r_2\to r_2]{r_3+r_1\to r_1} \begin{bmatrix} 1 & 0 & 5 & 0 & -9 & 1 \\ 0 & 1 & -1 & 0 & 5 & -10 \\ 0 & 0 & 0 & 1 & -1 & 7 \\ 0 & 0 & 0 & 0 & 1 & 8 \end{bmatrix} \xrightarrow[-5r_4+r_2\to r_2]{9r_4+r_1\to r_1} \begin{bmatrix} 1 & 0 & 5 & 0 & 0 & 73 \\ 0 & 1 & -1 & 0 & 0 & 50 \\ 0 & 0 & 0 & 1 & 0 & 15 \\ 0 & 0 & 0 & 0 & 1 & 8 \end{bmatrix}$

$r_3+r_4\to r_4$ $r_4+r_3\to r_3$

20. (C) $A^2(A^{-1}+B^{-1})B(A+B)^{-1}B = A(B+A)(B+A)^{-1}B = \boxed{AB}$ (C)

21. (a) $\begin{pmatrix} 2 & a & 1 \\ -3 & b & 3 \\ 1 & c & -2 \end{pmatrix} \xrightarrow{\frac{r_1}{2}\to r_1} \begin{pmatrix} 1 & \frac{a}{2} & \frac{-1}{2} \\ -3 & b & 3 \\ 1 & c & -2 \end{pmatrix} \xrightarrow[-r_1+r_3\to r_3]{3r_1+r_2\to r_2} \begin{pmatrix} 1 & \frac{a}{2} & \frac{-1}{2} \\ 0 & b+\frac{3a}{2} & \frac{3}{2} \\ 0 & c-\frac{a}{2} & \frac{-3}{2} \end{pmatrix}$ $r_2=-r_3$

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$\therefore b+\frac{3a}{2}=\frac{a}{2}-c \therefore \boxed{b+a+c=0}$ (a)

22. (b)

23. (d)

$\begin{bmatrix} 1 & -2 & 4 & a \\ 2 & -7 & 11 & b \\ 3 & -3 & 21 & c \\ 4 & 3 & 2 & d \end{bmatrix} \xrightarrow[-3r_1+r_3\to r_3]{-2r_1+r_2\to r_2} \begin{bmatrix} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b-2a \\ 0 & 3 & 9 & c-3a \\ 0 & 11 & -14 & d-4a \end{bmatrix} \xrightarrow[-11r_2+r_4\to r_4]{-3r_2+r_3\to r_3} \begin{bmatrix} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b-2a \\ 0 & 0 & 0 & c+3a-3b \\ 0 & 0 & -47 & d+18a-11b \end{bmatrix}$

$c+3a-3b=0$

$\boxed{c=3b-3a}$ (b)

$d+18a-11b=-47$

$\therefore \boxed{d=11b-18a-47}$ (d)

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